

WELL COMPLETION REPORT (SYNTHETIC)

OIL-MRN-004 - Moran-004

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000039
Field / District	Moran / Dibrugarh
Block	PEL-ASSAM-03
Basin	Assam-Arakan
Formation	Girujan (GRJ)
Spud / Completion	2015-10-05 / 2015-12-26
Measured / TVD	2328 m / 2306 m
Trajectory	Deviated; inc 17.0 deg; az 54.3 deg
Formation Interval	1259-1909 m
Lithology	Silty shale with sandstone lenses
Pore Pressure Reference	2403 psi
Fracture Gradient	13.86 ppg
Mud System	KCl Polymer WBM
NWIS Well Risk	Low (41.1/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	582	Surface
9 5/8 in casing	1350	Intermediate
7 in liner/casing	2048	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2015-10-21 577 m	Kick	Critical	Flow-out exceeded flow-in and gas units increased. Mitigation: Increase mud weight within fracture-margin limits and monitor gas trend. Lesson: Maintain mud-weight window using offset pressure evidence.
2015-11-04 980 m	NPT	High	No depth progress occurred during the recorded interval. Mitigation: Verify equipment health before resuming drilling. Lesson: Separate geological risk from equipment-driven NPT in analytics.
2015-11-05 994 m	Pressure Spike	High	Standpipe pressure increased rapidly at nearly stable flow. Mitigation: Condition hole and confirm pressure normalization. Lesson: Use depth-matched offset hydraulic baseline for early warning.
2015-10-19 534 m	Mud Loss	Medium	Intermittent circulation losses developed while drilling ahead. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2015-10-25 704 m	Torque Spike	Medium	Rotary speed fluctuated during constant pump rate. Mitigation: Monitor torque derivative rather than absolute torque alone. Lesson: Use offset torque trend to set adaptive operating envelope.
2015-11-04 966 m	Cementing Issue	Medium	Observed returns did not match calculated displacement. Mitigation: Review loss history before cementing equivalent interval. Lesson: Volume imbalance is an important cement-placement warning.
2015-11-11 1155 m	Kick	Medium	Drilling break followed by positive flow check. Mitigation: Maintain trip margin and use conservative connection practices. Lesson: Escalate connection-gas trends before entering correlated pressure zone.

Completion Summary

The synthetic well reached 2328 m MD in the Girujan interval. The NWIS knowledge record links 13 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.